DAY 22 | POST-READ

Embeddings and Retrieval: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Embeddings	Vectors that capture meaning; cosine similarity; distance = 1 - similarity; one model per index.
Vector database	Indexes, metadata, filters, updates and scale. FAISS, Chroma, pgvector, Qdrant, Azure AI Search.
Flat, HNSW, IVF	Flat is exact; HNSW (graph) and IVF (clusters) trade a little recall for big speed.
BM25	Keyword relevance; great for codes, names and rare terms.
Hybrid + RRF	Fuse ranked lists with sum of 1 / (k + rank).
Re-ranking	Score query-chunk pairs with a cross-encoder or LLM; keep the best few.
Measure	hit@k and MRR on a test set; refusal accuracy on unanswerable questions.
Choosing an approach (NEW v4)	RAG for large, changing facts; fine-tuning for style; long context for small corpora; caching for stable prefixes.

Part B: Going beyond today

1. Query rewriting

Ask the model to rewrite a vague or conversational question into a clear search query. "And for Japan?" becomes "roaming charges in Japan".

2. HyDE

Hypothetical Document Embeddings: ask the model to draft an answer, embed the draft, and search with it. Drafts often look more like the target passages than short questions do.

3. Cost-aware RAG

Retrieval top-k drives input tokens. Retrieve wide, re-rank, and send only the best 3 to 5 chunks. Day 25 goes deeper on token optimisation.

4. Re-indexing

When you change embedding model or chunking, re-embed everything. Mixing vectors from different models in one index gives meaningless similarities.

5. Combining approaches

Real systems mix methods: a cached system prompt and tool list, RAG for facts, and sometimes a fine-tuned or small model for classification.

Part C: Check yourself

- Why can two sentences with no shared words have similar embeddings?
- What does ANN give up in exchange for speed? How do HNSW and IVF differ?
- When does BM25 beat vector search?
- Why retrieve 20 candidates and re-rank to 3, instead of retrieving 3?
- A 600-page manual changes monthly and gets 10,000 questions a day. RAG, fine-tuning, long context or caching? Why?

Part D: Practice ideas for tonight

- Add metadata filtering by section to your retrieval function.
- Compute hit@3 and MRR for keyword, vector, hybrid and hybrid + re-rank on ten questions.
- Change the placeholder prices and query volume in the cost cell. When does caching win over RAG?

Coming up on Day 23: GenAI use cases by industry, maturity levels, and an SME session on security in the agentic world.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today